

Classic Riddler

18 March 2022

Riddle:

A postal worker and his customer joke about the various ways the customer could mathematically encode her post office box number.

The customer realizes that every integer greater than 1 can be encoded via at least one Fibonacci-like sequence using an ordered triple (m, n, q) . The encoded number is the q^{th} member of the sequence after the first two positive integers m and n , where each term is the sum of the previous two terms. For example, 7 has the encodings $(3, 4, 1)$ and $(1, 3, 2)$.

In an attempt to stump the postal worker, the customer prefers encodings with a maximal value of q . What encoding should she use for the number 81?

Extra credit: What encoding should she use for the number 179?

Solution:

For any Fibonacci-like sequence, the ratio of consecutive numbers will always approach the golden ratio, $\phi \approx 1.618$. Therefore, for the number 81 to be as far as possible along in the sequence, its ratio to the previous number should be close to this value. Specifically, $80/1.618 \approx 50.1$, so the best candidates for the next-to-last number are 50 and 51. Working backwards from 50, the reversed sequence would be 81, 50, 31, 19, 12, 7, 5, 2, 3. This is as far as it can go with positive integers to start. This has a q of 7. For 51, the reversed sequence would be 81, 51, 30, 21, 9, 12, which has a q of 4. So the solution to the first question is **(3,2,7)**.

Using the same reasoning for 179, $179/1.618 \approx 110.6$. With 110, the reversed sequence would be 179, 110, 69, 41, 28, 13, 15, which has a q of 5. With 111, the reversed sequence would be 179, 111, 68, 43, 25, 18, 7, 11, which has a q of 6. So the solution to the second question is **(11,7,6)**.

I wrote a table in `postal_sums.xlsx` that calculates the maximum q value for combinations of m and n up to 20 and 20. This confirms the manual solutions above.